THE UNIVERSITY OF AUCKLAND

SEMESTER ONE 2025
Campus: City

COMPUTER SCIENCE

COMPSCI 714: AI Architecture and Design

MID-SEMESTER TEST
30h of April 2025

Time Allowed: 60 minutes (5 minutes reading time + 55 min answering time)

NOTES:

- This test is out of **20** marks and counts towards **10%** of your final course grade.
- There are 6 short-answer questions. Attempt **ALL** questions.
- This is a restricted book test. You are only allowed to use a **double-sided page of handwritten notes**.
- Digital devices, including calculators and phones are not permitted.
- Write your answers directly after each question.
- Cross out notes and other work you do not wish to be assessed.
- We privilege quality over quantity, i.e., you do not need to write very long answers. Be concise and clear.
- Write as clearly as possible, in plain English.
- Overflow space is available at the end of the test paper, indicate at the end of the original question if you are using overflow space.

Question	Topic	Out of	Marks
1	Dataset cleaning	2	
2	Evaluation and DC	3	
3	Activation functions	3	
4	Learning rate	4	
5	Transformers	4	
6	CNNs	4	
TOTAL	/	**20**	

Name:	**UPI:**	**ID:**

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Answer ALL questions.

Question 1: Dataset cleaning [2 marks]

Consider the following summary of a dataset containing 5 attributes and 10 000 samples. The table gives the type and number of missing values for each attribute, as well as the mean, standard deviation, minimum and maximum value for each numerical attribute. The numerical attributes follow a normal distribution.

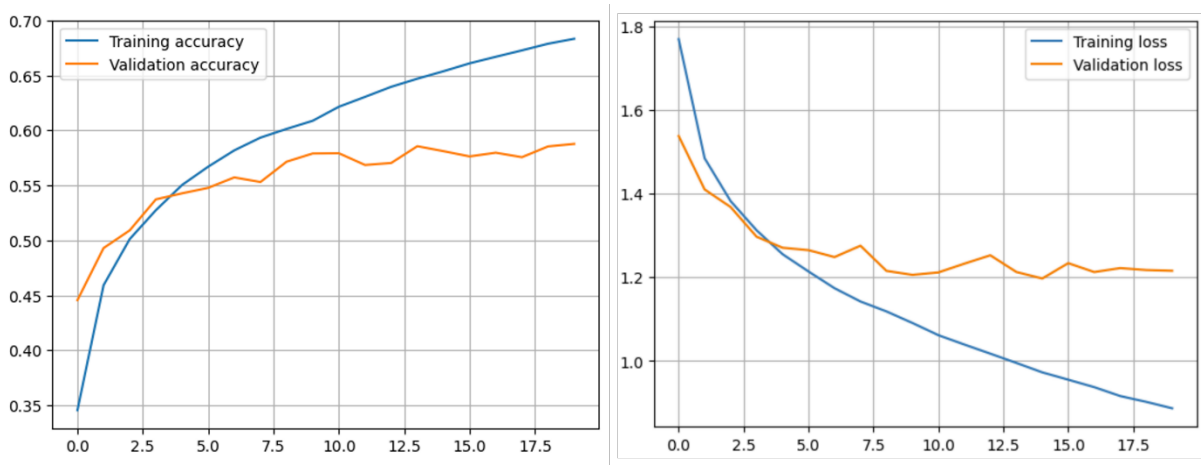
Attribute	Type	Missing values	Mean	Standard deviation	Max	Min
Attribute 1	Binary	0	/	/	/	/
Attribute 2	Numerical	15	500.5	103.1	50000.0	-1000.0
Attribute 3	Categorical	0	/	/	/	/
Attribute 4	Numerical	9995	1.2	0.2	2.0	0.0
Attribute 5	Numerical	23	25360.5	30215.2	125000.0	-75000.0

For each of the following cleaning steps, explain if it makes sense to apply it to this dataset. Briefly justify your answers. [0.5 mark per step]

- Missing value replacement based on most frequent value.
- Missing value replacement based on median value.
- Removing an attribute.
- Outlier removal.

Question 2: Evaluation and Design Choices [3 marks]

After training a neural network for 20 epochs, you get the following accuracy and cross entropy loss curves:



(a) Describe the model's performance in terms of bias and variance. Use the above plots to explain your answer. [1 mark]

Question continues on the next page.

(b) Suggest two different changes you could make to the model or the training process to improve on the previous validation performance. Each suggested change should be affecting a different aspect of the model or training process. Briefly explain the expected effect of each of your suggested change. [1 mark per suggestion]

Question 3: Activation functions [3 marks]

(a) LeakyReLU is a variation of the ReLU activation function and is defined by the following equation:

$$LeakyReLU(x) = \begin{cases} x, & x > 0 \\ \alpha x, & x \leq 0 \end{cases}$$

with α typically set to 0.3-0.01.

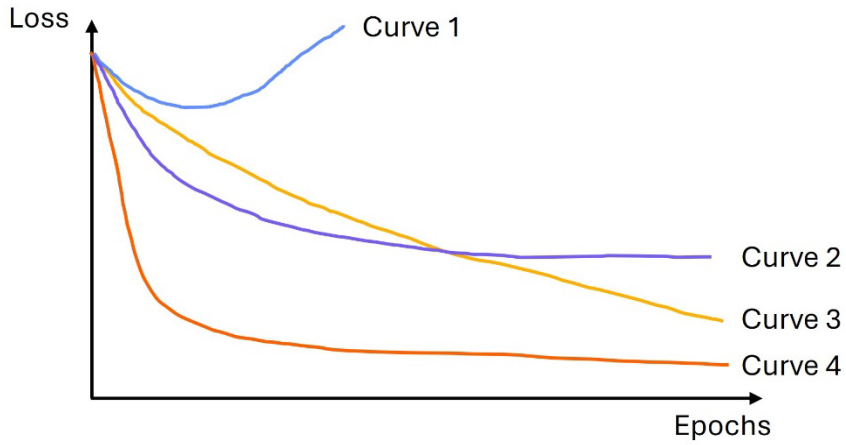
Explain what problem can arise when using ReLU, and how LeakyReLU helps mitigate it. [1 mark]

Question continues on the next page.

(b) You have been hired by a manufacturing company to design an automated visual quality control system using a neural network. The system analyses images of manufactured products and predicts whether various types of visual anomalies are present.
Note that a single image may contain multiple anomalies simultaneously. Based on this scenario, what activation function would you choose for the output layer of the network? Justify your choice briefly. [2 marks]

Question 4: Learning rate [4 marks]

Consider the 4 loss curves below.



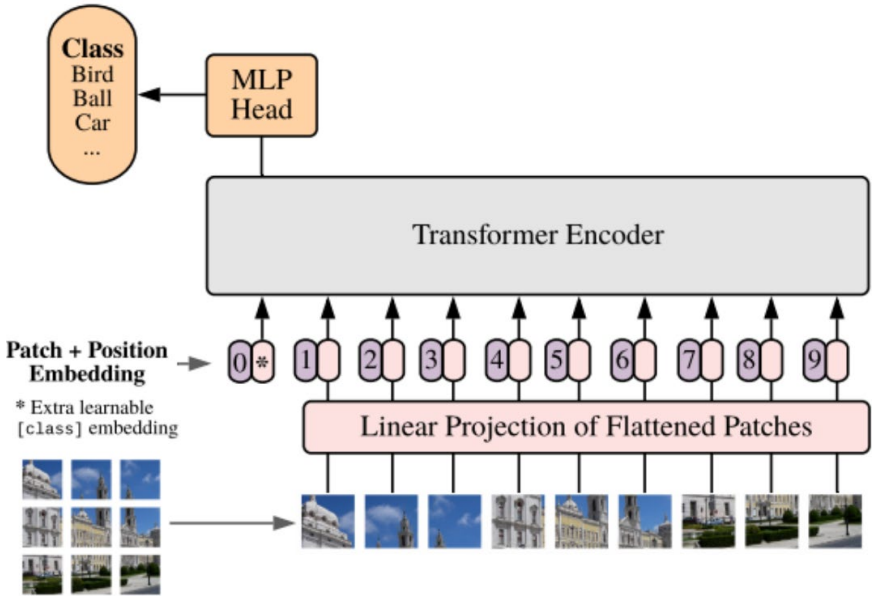
Each curve corresponds to a different learning rate from the following options: 0.5, 0.1, 0.01, and 0.001.
Based on the shape of each loss curve and your understanding of how learning rate affects training, match each curve to the most likely learning rate. Briefly justify your reasoning. [1 mark per curve]

Question 5: Transformers [4 marks]

(a) The Transformer decoder architecture includes a masked multi-head attention module. Briefly explain what the purpose of the masking is in this module. [2 mark]

Question continues on the next page.

(b) In the Vision Transformer architecture shown below, explain what the role of the extra learnable embedding added at the start of the input sequence is. Discuss briefly the main advantage of using this approach. [2 marks]



Question 6: CNNs [4 marks]

Consider the following CNN architecture:

- Input images' shape: [35, 35, 3]
- Convolutional layer 1: 10 filters with kernel size = 7, stride = 2 and valid padding.
- Max pooling layer 1: kernel size = 2 and stride = 2.
- Convolutional layer 2: 20 filters with kernel size = 3, stride = 1 and same padding.
- Max pooling layer 1: kernel size = 2 and stride = 2.
- Fully connected layer 1: **?** inputs and 10 outputs.

(a) What is the number of inputs to the fully connected layer? Choose 1 answer. [1 mark]

i. 9
ii. 180
iii. 980
iv. 20
v. 1280

(b) Show your workings. [3 marks]

END OF TEST QUESTIONS – OVERFLOW PAGES FOLLOW

OVERFLOW PAGES